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SOME STATISTICS ON A 701 HYDRODYNAMICS CODE

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The following counts of instructions of a typical Hydrodyn Code have been made with the hope that data of this kind may be helpf discussions of machine logic, etc.

There were two types of counts made: (1) The "Static" coun made by counting the number of orders present in the regional cards o high speed sorter. Utility programs, A-Regions, etc were not include (2) The "dynamic" count was made by going through a typical calculatio cycle and counting the numbers of orders actually executed in that cy

I have left out the input-output orders (Read, copy, etc.) they are only incidental to the main hydrodynamics code. A study of utility programs would be more pertinent for this purpose.

The "raw" data is included in Table I, since there are no d a large number of ways that it can be interpreted other than those gi in Tables II, III, and IV. The conclusions I draw (if any) are that must be careful in using static counts to infer dynamic behavior. Mo of the ratios in Table IV are very close except the important one, th add to mpy ratio.

TABLE I

ORIGINAL DATA

Order No.	Order	Static Count No. Orders	Dynamic Count No. Orders
00	Stop	61	0
01	Tr	133	27
02	Tr on	7	4
03	Tr plus	27	9
04	Tr zero	42	9
05	Sub	116	33
06	R Sub	23	10
09	Add	122	62
10	R Add	308	77
11	Add ab	5	7
12	Store	199	79
13	Store A	169	37
14	Store MQ	19	12
15	Load MQ	47	29
16	Mpy	32	22
17	Mpy R	44	33
18	Div	27	19
20	L Left	19	10
21	L Rt.	36	26
22	A Left	13	4
23	A Rt.	7	0
		total = 1456 orders	total = 509 orders

213. 214

243

102.

TABLE II

PERCENTAGES ACCORDING TO ORDER TYPE

Order Type	Static Count		Dynamic Count	
	no. orders	percent	no. orders	percen
Transfers	270	18.5	49	9.6
Adds, Subs, LMQ	765, 621	42.7	244, non carry adds (fx.) 218	42.8
Store	243, 387	26.6	102 adds (fl.) 128	25.1
Mpys	76	5.2	55	10.8
Div	27	1.9	19	3.7
Shifts	75	5.2	40	7.9
	1456		509	

TABLE III

PERCENTAGES WITH ATTEMPT TO REMOVE EFFECT OF INDEX REGISTERS.
(STORE ADDRESS ORDERS ARE REMOVED ALONG WITH ONE ADD ORDER EACH.)

Order Type	Static Count		Dynamic Count	
	No. orders	percent	No. orders	percen
Transfers	270	23.6	49 adds (fx)	11.3
Adds, Subs, LMQ	427, 452	39.5	170, non-carry adds 181	41.6
Stores	243, 218	19.1	102 adds (fl.) 91	20.9
Mpys	76	6.6	55	12.6
Div	27	2.3	19	4.3
Shifts	75	6.6	40	9.2
	1143		435	

TABLE IV

SOME RATIOS OF INTEREST (TAKEN FROM TABLE III)

Ratio	Static	Dynamic
Adds per Mpy	5.9	3.3
Mpys per Div	2.8	2.9
Stores per add	0.48	0.50
Shifts per Div	2.8	2.1
fixed adds / Mpy	5.6	3.1
floating adds / Mpy	3.2	1.85

approx: (fx. adds) : (fl. adds) : (Mpy) : (Div) = 6 : 4 : 3 : 1